

1. Workload

2. ISA & ABI

3. Compute Org

4. Memory & NoC

5. Power Intent

6. Compiler/RT

7. Physical Node

8. Verification

■ □ CHAPTER STACK SCOPE — FULL STACK ACTIVE (PATTERNS 1-4)
□ ■ Loop Patterns Across the System Stack

1. Pattern Catalog

- Spec-to-RTL Refinement
- Surrogate-Accelerated DSE
- Verification Repair Loop
- Physical P&R Tuning

2. Stack Alignment

- Workload to Layout Rules
- Cross-Layer Invariants
- Composition Contracts
- Interface Standardization

3. Pattern Metrics

- Candidate Quality Yield
- Verification Speedup Ratio
- Compute Cost Efficiency
- Human Review Overhead

4. Architecture Guide

- Pattern Selection Matrix
- Anti-Pattern Warnings
- Production Playbook
- Handoff Blueprint

Pattern Optimization & Playbook Refinement Loop